Q1

1.

```
. reg wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	61,136
				F(2, 61133)	=	4318.10
Model	3.2226e+14	2	1.6113e+14	Prob > F	=	0.0000
Residual	2.2812e+15	61,133	3.7315e+10	R-squared	=	0.1238
				Adj R-squared	=	0.1238
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17826.42	193.9603	91.91	0.000	17446.26	18206.59
nin	-6880.068	714.7795	-9.63	0.000	-8281.038	-5479.099
_cons	72117.21	2519.881	28.62	0.000	67178.24	77056.19

$\beta_1 = 17826.42$ s.e. $\beta_1 = 193.9603$

$\beta_2 = -6880.068$ s.e. $\beta_2 = 714.7795$

the p-values are 0.000. which means

$H_0\ \beta_1 = 0$ and $H_0\ \beta_2 = 0$ can both be rejected at 1% significance level

$R^2 = 0.1238$

About 12.38% is the variation in wage for these individuals is explained by their education years and nin.

2.

```
. reg wage yeduc
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	8530.77
Model	3.1880e+14	1	3.1880e+14	Prob > F	=	0.0000
Residual	2.2846e+15	61,134	3.7371e+10	R-squared	=	0.1225
				Adj R-squared	=	0.1224
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17910.03	193.9109	92.36	0.000	17529.96	18290.09
_cons	57405.06	2004.897	28.63	0.000	53475.45	61334.66

$\tilde{\beta}_1 = 17910.03$ $\quad \tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \delta_1$

bias $= \hat{\beta}_2 \delta_1 = 83.61$ it is possitive direction

3. $\delta_1 = \dfrac{cov(yeduc, nin)}{Var(yeduc)}$ $\quad \hat{\beta}_2 < 0$

$\therefore \delta_1 < 0 \quad \therefore cov(yeduc, nin) < 0$

$\therefore$ it is negative

Based on the data set

```
. reg nin yeduc
```

Source	SS	df	MS	Number of obs	=	152,989
				F(1, 152987)	=	592.60
Model	762.477939	1	762.477939	Prob > F	=	0.0000
Residual	196843.137	152,987	1.28666578	R-squared	=	0.0039
				Adj R-squared	=	0.0039
Total	197605.615	152,988	1.29164127	Root MSE	=	1.1343

nin	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	-.017078	.0007015	-24.34	0.000	-.018453	-.015703
_cons	2.24942	.0063333	355.17	0.000	2.237006	2.261833

δ_1 is also < 0 so they are consistent

4

```
. reg wage yeduc nin en71
```

Source	SS	df	MS	Number of obs	=	60,936
				F(3, 60932)	=	2873.13
Model	3.2216e+14	3	1.0739e+14	Prob > F	=	0.0000
Residual	2.2774e+15	60,932	3.7376e+10	R-squared	=	0.1239
				Adj R-squared	=	0.1239
Total	2.5996e+15	60,935	4.2661e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17704.64	197.781	89.52	0.000	17316.98	18092.29
nin	-7513.557	735.1522	-10.22	0.000	-8954.457	-6072.656
en71	35557.16	9684.733	3.67	0.000	16575.06	54539.27
_cons	68233.96	2736.359	24.94	0.000	62870.69	73597.23

$wage_i = \beta_0 + \beta_1 yeduc_i + \beta_2 nin_i + \beta_3 en71_i + \mu$

the en71 also change the wage outcome.

5. ① $H_0 : \beta_3 = 0$ H_1 $\beta_3 \neq 0$

② $t = \frac{\hat{\beta}_3}{s.e.\beta_3} = \frac{35557.16}{9684.733} = 3.67$

③ $\alpha = 1\%$, two sided

④ $3.67 > 2.576$ $= C_{99.5\%}$

⑤ so the H_0 can be rejected

at significance level of 1%

6. use F - test

$H_0: \beta_2 = \beta_3 = 0$

unrestricted model

$wage_i = \beta_0 + \beta_1 yeduc_i + \beta_2 nin_i + \beta_3 en71_i + u$

restricted model

$wage = \beta_0 + \beta_1 yeduc + u$

$$F = \frac{(SSR_r - SSR_u)/q}{SSR_u/(n-k-1)} \sim F(q, n-k-1)$$

choose $\alpha = 1\%$

```
. test nin en71

 ( 1)  nin = 0
 ( 2)  en71 = 0

       F(  2, 60932) =   53.17
            Prob > F =    0.0000
```

p-value = 0.0000 < 1%

we can reject the H_0 at significance level of 1%

Q2.

1. 1) linear

```
. reg high_wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	152,989
				F(2, 152986)	=	39.17
Model	12.4959378	2	6.24796889	Prob > F	=	0.0000
Residual	24399.7853	152,986	.159490315	R-squared	=	0.0005
				Adj R-squared	=	0.0005
Total	24412.2812	152,988	.159569909	Root MSE	=	.39936

high_wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	.0013518	.0002475	5.46	0.000	.0008668	.0018368
nin	.0065628	.0009001	7.29	0.000	.0047985	.008327
_cons	.7760058	.0030119	257.64	0.000	.7701025	.7819091

2) Probit model

```
. probit high_wage yeduc nin

Iteration 0:  Log likelihood = -76403.605
Iteration 1:  Log likelihood = -76363.713
Iteration 2:  Log likelihood = -76363.703
Iteration 3:  Log likelihood = -76363.703

Probit regression                                 Number of obs = 152,989
                                                  LR chi2(2)    =   79.80
                                                  Prob > chi2   =  0.0000
Log likelihood = -76363.703                       Pseudo R2     =  0.0005
```

high_wage	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
yeduc	.0049024	.0008971	5.46	0.000	.0031441	.0066606
nin	.0241636	.0033068	7.31	0.000	.0176824	.0306447
_cons	.7542664	.0108561	69.48	0.000	.7329888	.775544

Table :

high_wage = 1 if wage ≥ 19200, = 0 if wage < 19200

reg model	LPM	probit
yeduc	0.0489753	0.1370616
	(0.0034608)	(0.0014279)
nin	-0.0012381	-0.0346085
	(0.0001681)	(0.0004821)
constant	0.0600217	-1.233631
	(0.0058863)	(0.1785)

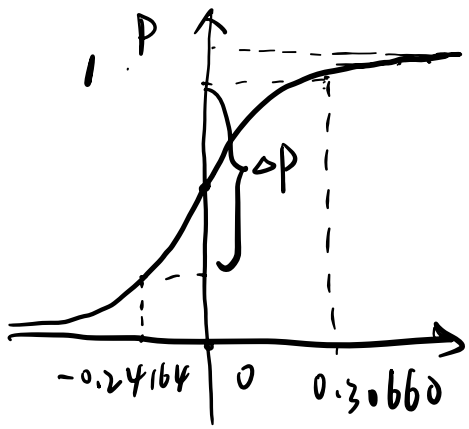
the coefficient of yeduc in LPM means that one additional year of education will increase probability of high wage with 4.89753%

2. ① probably larger than 1 and less than 2
② every one additional unit of independent variable has constant effect of the dependent
③ Violate the homoskedasticity

3. $\Delta p = \Phi(\beta_0 + \beta_1 \times 12 + \beta_2 \times 3) - \Phi(\beta_0 + \beta_1 \times 8 + \beta_2 \times 3)$

$= \Phi(0.30660) - \Phi(-0.24164)$

4.



Q3

1.

```
. reg wage high_level
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	4561.41
Model	1.8076e+14	1	1.8076e+14	Prob > F	=	0.0000
Residual	2.4227e+15	61,134	3.9629e+10	R-squared	=	0.0694
				Adj R-squared	=	0.0694
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	2.0e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
high_level	112764.3	1669.638	67.54	0.000	109491.8	116036.8
_cons	156631.9	1327.543	117.99	0.000	154029.9	159233.9

people receiving education over or equal to 9 years have 112764.3 more in monthly wage than those who receives less than 9 years

2. $wage = \beta_0 + \beta_1 high_level + \beta_2 urban + \beta_3 urban\,high_level + u$

```
. reg wage high_level urban urbanhigh_level
```

Source	SS	df	MS	Number of obs	=	61,136
				F(3, 61132)	=	1727.11
Model	2.0342e+14	3	6.7805e+13	Prob > F	=	0.0000
Residual	2.4000e+15	61,132	3.9259e+10	R-squared	=	0.0781
				Adj R-squared	=	0.0781
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	2.0e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
high_level	99802.76	1914.016	52.14	0.000	96051.29	103554.2
urban	25014.22	3553.394	7.04	0.000	18049.56	31978.88
urbanhigh_level	23411.8	4131.893	5.67	0.000	15313.28	31510.32
_cons	152485.9	1446.661	105.41	0.000	149650.4	155321.3

$\beta_0 = 152485.9$

$\beta_1 = 99802.78$

$\beta_2 = 25014.22$

$\beta_3 = 23411.8$

2) $H_0 : \beta_2 = \beta_3 = 0$

restricted function is

$wage = \beta_0 + \beta_1 high_level + u$

$$F = \frac{(SSR_r - SSR_u)/q}{SSR_u/(n-k-1)} = 288.51 \sim F(2, 61132)$$

choose $\alpha = 1\%$

p-value = 0.000 < 1%

∴ we can reject the Ho at significance level of 1%

So urban can affect the slope of high_level

do-file

```
clear

cap log close

log using ps2.log,  replace

cd"/Users/hanqihang/Stata/PS2"

//loding the data
use inpresdata.dta

//model 1
reg wage yeduc nin

//model 2
reg wage yeduc

//check the corr of nin and yeduc
reg nin yeduc

//model 3
reg wage yeduc nin en71

//find the joint significance of nin and en71
test nin en71

gen high_wage = (wage >= 192000)

//linear model
reg high_wage yeduc nin

//probit model
probit high_wage yeduc nin

//generate high_educ
gen high_level = (yeduc >= 9)

//show the relationship between wage and high level
reg wage high_level

//generate urbanhigh_level
gen urbanhigh_level = ( urban == 1 & high_level == 1)

//regess in the interaction term
reg wage high_level urban urbanhigh_level

//check if the urban will affect the slope of high_level
test urban urbanhigh_level

//close the log
log close
```